Shiqi Liu

shiqiliu@andrew.cmu.edu | linkedin.com/in/shiqiliu2 | shiqiliu-67.github.io

Research Profile

Ph.D. candidate at Carnegie Mellon University developing robot learning methods that help robots adapt to new tasks and environments. Experience spans reinforcement and imitation learning, geometric learning, agentic systems, and sim-to-real deployment on robotic arms and quadrupedal manipulators.

Education

Carnegie Mellon University **May 2024 – Present**
Doctor of Philosophy, Mechanical Engineering; GPA: 4.00/4.00 *Pittsburgh, PA*

Carnegie Mellon University **Feb. 2021 – Dec. 2022**
Master of Science, Mechanical Engineering; GPA: 3.96/4.00 *Pittsburgh, PA*

Beijing University of Technology **Sept. 2014 – May 2018**
Bachelor of Energy and Power Engineering; GPA: 3.88/4.00 *Beijing, China*

Work Experience

Mitsubishi Electric Research Laboratories (MERL) **May – Aug. 2026**
Research Intern *Cambridge, MA*

- Developed ELEVATE, a system that enables a quadruped with an arm to learn missing physical skills from task failures and reuse them for later manipulation tasks.
- Built a Python/Isaac Lab workflow combining multimodal simulation feedback with agent-directed reinforcement learning using supplied PPO infrastructure; expanded the controller library from 1 to 4 and achieved 92%–100% simulation success on seven tasks, with 50 held-out variants per task.
- Deployed controllers on a Unitree Go2 with an AgileX PiPER arm and image-based target localization, achieving 18/20 rack-retrieval and 16/20 floor-pickup/rack-placement successes without real-robot policy fine-tuning.

Selected Publications

* Equal contribution.

1. **Shiqi Liu**, Radu Corcodel, and Ding Zhao. ELEVATE: Agentic Controller Acquisition from Task Failure for Quadrupedal Manipulation. *Submitted to ICRA 2027.*
2. **Shiqi Liu**, Yihang Yao, Peide Huang, Mengdi Xu, and Ding Zhao. MIRROR: Reflecting Hidden Symmetries from Offline Robot Interaction Data. *Conference on Robot Learning (CoRL)*, 2026.
3. Yikai Wang*, Tingxuan Leng*, Changyi Lin*, **Shiqi Liu**, Shir Simon, Bingqing Chen, Jonathan Francis, and Ding Zhao. APEX: Learning Adaptive High-Platform Traversal for Humanoid Robots. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
4. Xilun Zhang*, **Shiqi Liu***, Peide Huang, William Jongwon Han, Yiqi Lyu, Mengdi Xu, and Ding Zhao. Dynamics as Prompts: In-Context Learning for Sim-to-Real System Identifications. *IEEE Robotics and Automation Letters (RA-L)*, 2025.
5. Changyi Lin, Yuxin Ray Song, Boda Huo, Mingyang Yu, Yikai Wang, **Shiqi Liu**, Yuxiang Yang, Wenhao Yu, Tingnan Zhang, Jie Tan, Yiyue Luo, and Ding Zhao. LocoTouch: Learning Dynamic Quadrupedal Transport with Tactile Sensing. *Conference on Robot Learning (CoRL)*, 2025.
6. Ziyi Xu*, Haohong Lin*, **Shiqi Liu***, and Ding Zhao. Query-Centric Diffusion Policy for Generalizable Robotic Assembly. *arXiv preprint arXiv:2509.18686*, 2025.

7. **Shiqi Liu**[*], Mengdi Xu[*], Peide Huang, Xilun Zhang, Yongkang Liu, Kentaro Oguchi, and Ding Zhao. Continual Vision-based Reinforcement Learning with Group Symmetries. *Conference on Robot Learning (CoRL)*, 2023. **Oral presentation**.
8. Peide Huang, Xilun Zhang[*], Ziang Cao[*], **Shiqi Liu**[*], Mengdi Xu, Wenhao Ding, Jonathan Francis, Bingqing Chen, and Ding Zhao. What Went Wrong? Closing the Sim-to-Real Gap via Differentiable Causal Discovery. *Conference on Robot Learning (CoRL)*, 2023.
9. Mengdi Xu[*], Peide Huang[*], Wenhao Yu[*], **Shiqi Liu**, Xilun Zhang, Yaru Niu, Tingnan Zhang, Fei Xia, Jie Tan, and Ding Zhao. Creative Robot Tool Use with Large Language Models. *CoRL Workshop on Language and Robot Learning: Language as Grounding*, 2023.

Selected Research Projects

MIRROR: Learning Symmetries for Visual Manipulation **Aug. 2025 – May 2026**
Carnegie Mellon University, Safe AI Lab | First author; CoRL 2026

- Developed a method to learn how robot observations and actions transform together across equivalent task configurations, enabling reuse of offline interaction data for imitation learning.
- Implemented adversarial symmetry learning with group-consistency constraints in PyTorch; integrated learned transformations for a specified rotation group into behavior cloning, raising mean RGB simulation success from 80.48% to 85.12% across four manipulation tasks.
- Built a Kinova GEN3 deployment pipeline with calibrated RGB-D heightmaps and SAM 2 masks; transferred policies trained for these inputs in simulation to four physical tasks without real-world policy fine-tuning.

CAPTURE: In-Context Sim-to-Real System Identification **May – Dec. 2024**
Carnegie Mellon University, Safe AI Lab | First author (equal contribution); RA-L 2025

- Co-developed a method that uses recent robot interactions to adapt simulator physics without updating model weights at deployment, helping transfer learned skills to real hardware.
- Co-developed the code and identified binary-search parameter-update histories as effective training data for iterative dynamics adaptation; designed scooping and air-hockey tasks for simulation and physical evaluation.
- Set up and ran physical experiments, achieving scooping success of 90% for eggplant, 70% for celery, and 90% for carrot after seven adaptation iterations, with ten trials per object.

QDP: Vision-Language Planning for Robotic Assembly **May – Sept. 2024**
Carnegie Mellon University, Safe AI Lab | First author (equal contribution); 2025 preprint

- Contributed to a robotic furniture-assembly system that connects high-level decisions about parts and skills to a low-level diffusion policy for precise manipulation.
- Built the high-level sequence generator using SAM 2 segmentation, annotated visual prompts, and interaction history to select skills, components, and contact points and replan after execution.

COVERS: Continual Robot Learning through Symmetry **May 2022 – May 2023**
Carnegie Mellon University, Safe AI Lab | First author (equal contribution); CoRL 2023, oral

- Developed a method that helps robots learn new tasks while retaining prior skills by recognizing equivalent task configurations and reusing learned policies.
- Implemented equivariant networks and Wasserstein-distance task grouping to select and reuse PPO policies, achieving 91% mean simulation success versus 44% for 3RL across four groups of four Meta-World tasks, averaged over five runs.

Technical Skills

Programming and learning: Python, PyTorch.

Robotics and simulation: ROS, ROS 2, Isaac Lab, Isaac Sim, MuJoCo.

Robot platforms: Kinova GEN3, Unitree Go1/Go2/G1, AgileX PiPER, xArm.